

Time & Work — Answer Key & Solutions

Questions 21 – 32 | CAT / MBA Entrance Level

Question 21

A tank has an inlet pipe and an outlet pipe. If the outlet pipe is closed then the inlet pipe fills the empty tank in 8 hours. If the outlet pipe is open then the inlet pipe fills the empty tank in 10 hours. If only the outlet pipe is open then in how many hours will the full tank becomes half-full?

- A) 20
- B) 30
- C) 40
- D) 45

✓ **Correct Answer: A) 20**

Let I be the rate of the inlet pipe and O be the rate of the outlet pipe (in tanks per hour).

1. Inlet pipe alone fills in 8 hours $\rightarrow I = 1/8$
2. Both pipes open fills in 10 hours $\rightarrow I - O = 1/10$
3. Find O: $O = 1/8 - 1/10 = (5-4)/40 = 1/40$

The outlet pipe alone can empty a full tank in 40 hours.

4. To make the full tank half-full using only the outlet pipe:

Time = $(V/2) / O = (V/2) / (V/40) = 20$ hours.

Final Answer: 20 hours. Correct option: A

Question 22

Amal can complete a job in 10 days and Bimal can complete it in 8 days. Amal, Bimal and Kamal together complete the job in 4 days and are paid a total amount of Rs 1000 as remuneration. If this amount is shared by them in proportion to their work, then Kamal's share, in rupees, is

- A) 100
- B) 200
- C) 300
- D) 400

✓ **Correct Answer: A) 100**

Total work = LCM(10, 8, 4) = 40 units.

Amal's rate = $40/10 = 4$ units/day

Bimal's rate = $40/8 = 5$ units/day

Combined rate (A+B+K) = $40/4 = 10$ units/day

Kamal's rate = $10 - (4+5) = 1$ unit/day

Ratio of work = 4 : 5 : 1 $\rightarrow$ sum = 10

Kamal's share = $(1/10) \times 1000 = \text{Rs } 100$

Correct option: A

Question 23

A person can complete a job in 120 days. He works alone on Day 1. On Day 2, he is joined by another person who also can complete the job in exactly 120 days. On Day 3, they are joined by another person of equal efficiency. Like this, every day a new person with the same efficiency joins the work. How many days are required to complete the job?

[Type in the Answer — numerical]

✓ **Correct Answer: 15**

Each person does $1/120$ of the job per day. Total work = 120 units.

Day n has n people working $\rightarrow n$ units done.

Total work after n days = $1+2+3+\dots+n = n(n+1)/2$

Set $n(n+1)/2 = 120 \rightarrow n(n+1) = 240 \rightarrow 15 \times 16 = 240 \rightarrow n = 15$

Final Answer: 15 days

Question 24

Humans and robots can both perform a job but at different efficiencies. Fifteen humans and five robots working together take thirty days to finish the job, whereas five humans and fifteen robots working together take sixty days to finish it. How many days will fifteen humans working together (without any robot) take to finish it?

- A) 45
- B) 36
- C) 32
- D) 40

✓ **Correct Answer: C) 32**

Let h = human rate, r = robot rate (jobs per day).

Eq 1: $(15h + 5r) \times 30 = 1 \rightarrow 15h + 5r = 1/30$

Eq 2: $(5h + 15r) \times 60 = 1 \rightarrow 5h + 15r = 1/60$

Multiply Eq1 by 3: $45h + 15r = 1/10$

Subtract Eq2: $40h = 1/10 - 1/60 = 5/60 = 1/12 \rightarrow h = 1/480$

15 humans per day = $15/480 = 1/32$

Time = 32 days

Correct option: C

Question 25

When they work alone, B needs 25% more time to finish a job than A does. They two finish the job in 13 days in the following manner: A works alone till half the job is done, then A and B work together for four days, and finally B works alone to complete the remaining 5% of the job. In how many days can B alone finish the entire job?

- A) 20
- B) 22
- C) 16
- D) 18

✓ **Correct Answer: A) 20**

Let $r_A = 5x$, $r_B = 4x$ (since $T_B = 1.25 T_A$, rates are inversely proportional).

Stage 1: A alone does $W/2$.

Stage 2: A and B together do $0.45W$ in 4 days $\rightarrow (5x+4x) \times 4 = 36x = 0.45W \rightarrow W = 80x$.

Time for B alone = $W / r_B = 80x / 4x = 20$ days.

Correct option: A

Question 26

At their usual efficiency levels, A and B together finish a task in 12 days. If A had worked half as efficiently as she usually does, and B had worked thrice as efficiently as he usually does, the task would have been completed in 9 days. How many days would A take to finish the task if she works alone at her usual efficiency?

- A) 36
- B) 24
- C) 18
- D) 12

✓ **Correct Answer: C) 18**

Let A's rate = a , B's rate = b .

Condition 1: $(a+b) \times 12 = W \rightarrow 12a+12b = W \dots(1)$

Condition 2: $(a/2+3b) \times 9 = W \rightarrow 4.5a+27b = W \dots(2)$

From (1)=(2): $12a+12b = 4.5a+27b \rightarrow 7.5a = 15b \rightarrow a = 2b$

$W = 12(2b+b) = 36b$

Time for A alone = $W/a = 36b/(2b) = 18$ days.

Correct option: C

Question 27

John gets Rs 57 per hour of regular work and Rs 114 per hour of overtime work. He works altogether 172 hours and his income from overtime hours is 15% of his income from regular hours. Then, for how many hours did he work overtime?

[Type in the Answer — numerical]

✓ **Correct Answer: 12**

Let R = regular hours, O = overtime hours. $R + O = 172$.

Income from overtime = $O \times 114 = 0.15 \times R \times 57$

$114O = 8.55R \rightarrow 2O = 0.15R \rightarrow R = (40/3)O$

Substitute: $(40/3)O + O = 172 \rightarrow (43/3)O = 172 \rightarrow O = 172 \times 3/43 = 12$

Final Answer: 12 hours

Question 28

Three men and eight machines can finish a job in half the time taken by three machines and eight men to finish the same job. If two machines can finish the job in 13 days, then how many men can finish the job in 13 days?

[Type in the Answer — numerical]

✓ **Correct Answer: 13**

Let m = work rate of one man, c = work rate of one machine.

Time for 3 men & 8 machines = half time for 3 machines & 8 men:

$(3c+8m) \times t = W$ and $(8c+3m) \times (t/2) = W$

Equating: $3c+8m = (8c+3m)/2 \rightarrow 6c+16m = 8c+3m \rightarrow 13m = 2c$

Work rate of 2 machines = $2c = 13m$ = work rate of 13 men.

Since 2 machines finish in 13 days, 13 men also finish in 13 days.

Final Answer: 13 men

Question 29

A tank is fitted with pipes, some filling it and the rest draining it. All filling pipes fill at the same rate, and all draining pipes drain at the same rate. The empty tank gets completely filled in 6 hours when 6 filling and 5 draining pipes are on, but this time becomes 60 hours when 5 filling and 6 draining pipes are on. In how many hours will the empty tank get completely filled when one draining and two filling pipes are on?

[Type in the Answer — numerical]

✓ **Correct Answer: 10**

Let f = fill rate, d = drain rate per pipe.

Eq1: $(6f-5d) \times 6 = 1 \rightarrow 6f-5d = 1/6$

Eq2: $(5f-6d) \times 60 = 1 \rightarrow 5f-6d = 1/60$

Multiply Eq1 by 6 and Eq2 by 5:

$36f-30d = 1$; $25f-30d = 1/12$

Subtract: $11f = 11/12 \rightarrow f = 1/12$

From Eq1: $6(1/12)-5d = 1/6 \rightarrow 1/2-5d = 1/6 \rightarrow 5d = 1/3 \rightarrow d = 1/15$

Net rate (2 fill, 1 drain) = $2(1/12)-1(1/15) = 1/6-1/15 = (5-2)/30 = 1/10$

Time = 10 hours.

Question 30

Anil alone can do a job in 20 days while Sunil alone can do it in 40 days. Anil starts the job, and after 3 days, Sunil joins him. Again, after a few more days, Bimal joins them and they together finish the job. If Bimal has done 10% of the job, then in how many days was the job done?

- A) 12
- B) 13
- C) 15
- D) 14

✓ **Correct Answer: B) 13**

Total work = LCM(20,40) = 40 units.

Anil = 2 units/day, Sunil = 1 unit/day.

Bimal's work = 10% of 40 = 4 units.

Work by Anil & Sunil = 40 - 4 = 36 units.

Phase 1 (Anil alone, 3 days): $3 \times 2 = 6$ units.

Phase 2 (Anil+Sunil): need $36 - 6 = 30$ units at 3 units/day = 10 days.

Total days = 3 + 10 = 13 days.

Correct option: B

Question 31

Pipes A and C are fill pipes while Pipe B is a drain pipe of a tank. Pipe B empties the full tank in one hour less than the time taken by Pipe A to fill the empty tank. When pipes A, B and C are turned on together, the empty tank is filled in two hours. If pipes B and C are turned on together when the tank is empty and Pipe B is turned off after one hour, then Pipe C takes another one hour and 15 minutes to fill the remaining tank. If Pipe A can fill the empty tank in less than five hours, then the time taken, in minutes, by Pipe C to fill the empty tank is

[Type in the Answer — numerical]

✓ **Correct Answer: 90**

Let t_A = time for A, $t_B = t_A - 1$, t_C = time for C.

Rates: $A = 1/t_A$, $B = -1/(t_A - 1)$, $C = 1/t_C$.

Condition 1 (A,B,C together fill in 2 h):

$$1/t_A - 1/(t_A - 1) + 1/t_C = 1/2 \dots (1)$$

Condition 2 (B+C run 1 h, then C alone 1.25 h fills tank):

$$-1/(t_A - 1) + 1/t_C + 1.25/t_C = 1 \rightarrow -1/(t_A - 1) + 2.25/t_C = 1 \dots (2)$$

From (2): $1/(t_A-1) = 2.25/t_C - 1$

Substitute into (1): $1/t_A - (2.25/t_C - 1) + 1/t_C = 1/2$

$\rightarrow 1/t_A - 1.25/t_C + 1 = 1/2 \rightarrow 1/t_A = 1.25/t_C - 1/2 \dots(3)$

From (2): $t_C = 2.25(t_A-1)/t_A$ (one expression)

From (3): $t_C = 2.5t_A/(t_A+2)$ (another expression)

Equating $\rightarrow t_A^2 - 9t_A + 18 = 0 \rightarrow (t_A-3)(t_A-6)=0$

Since $t_A < 5$, $t_A = 3$ h.

$t_C = 2.25 \times (3-1)/3 = 2.25 \times 2/3 = 1.5$ h = 90 minutes.

Final Answer: 90 minutes

Question 32

Sam can complete a job in 20 days when working alone. Mohit is twice as fast as Sam and thrice as fast as Ayna in the same job. They undertake a job with an arrangement where Sam and Mohit work together on the first day, Sam and Ayna on the second day, Mohit and Ayna on the third day, and this three-day pattern is repeated till the work gets completed. Then, the fraction of total work done by Sam is

- A) $1/20$
- B) $3/10$
- C) $1/5$
- D) $3/20$

✓ **Correct Answer: B) $3/10$**

Sam = 20 days $\rightarrow$ 3 units/day (LCM=60 total work)

Mohit = 10 days $\rightarrow$ 6 units/day

Ayna = 30 days $\rightarrow$ 2 units/day

3-day cycle work:

Day 1 (Sam+Mohit): 9 units

Day 2 (Sam+Ayna): 5 units

Day 3 (Mohit+Ayna): 8 units

Total per cycle: 22 units

2 full cycles (6 days) = 44 units. Remaining = 16 units.

Day 7 (Sam+Mohit): 9 units done $\rightarrow$ remaining 7

Day 8 (Sam+Ayna): 5 units done $\rightarrow$ remaining 2

Day 9 (Mohit+Ayna): 2 units done $\rightarrow$ job complete

Sam works days 1,2,4,5,7,8 = 6 days $\times$ 3 = 18 units

Fraction = $18/60 = 3/10$

Correct option: B
